

ELECTRIC CIRCUITS

PART 2 MEASURING AND COMBINING

LESSONS 5–7

The resistance practical, and the rules for series and parallel circuits. Learn with it in class and revise from it at home.

- 5. Resistance of a wire
- 6. Series and parallel circuits
- 7. Resistors in series and parallel



NAME

CLASS

TEACHER

TARGET GRADE

HOW THIS BOOKLET WORKS

This is **Part 2** of the Electric Circuits workbook. It builds on the parts before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep it: it is your classwork book *and* your revision guide.

 DO NOW Quick questions from earlier lessons. Remembering again is what makes it stick.	 LEARN IT The explanation, diagrams and worked examples. Read this again when you revise.	 TRY IT Practice that builds up: first with support, then on your own.
 LAB Record measurements from a virtual lab or a required practical.	 EXAM CORNER An AQA-style question with a tip on how the marks are given.	 REVISE IT Summary, cover-and-answer questions and an “I can” check to colour in.

 **Revising at home?** For each lesson: (1) read the Learn it pages, (2) cover them and answer the “Cover the page” questions in Revise it, (3) check with the quick answers at the back, (4) do the matching mission on the companion website, which has more questions with answers.

THE ROUTE THROUGH THE UNIT · THIS PART IN BOLD

#	LESSON	COMPANION WEBSITE
01	Complete circuits and symbols What a circuit needs, circuit symbols and drawing rules, measuring current	Mission 1 · Build & measure
02	Current and charge Current as a rate of flow of charge, $Q = I t$, current is not used up	Mission 2 · Charge on the move
03	Potential difference p.d. as energy per coulomb, voltmeters, $E = Q V$, p.d.s add in series	Mission 3 · Energy per coulomb
04	Resistance and $V = I R$ Resistance, $V = I R$, ohmic conductors, why heating raises resistance	Mission 4 · Resistance & $V = I R$
05	Resistance of a wire REQUIRED PRACTICAL Required practical: how the length of a wire affects its resistance	Required practical · Resistance
06	Series and parallel circuits Current, p.d. and resistance rules for series and parallel circuits	Mission 5 · Series & parallel
07	Resistors in series and parallel REQUIRED PRACTICAL Required practical: measure total resistance of two resistors in series and in parallel	Required practical · Resistance (Part B)
08	I–V graphs I–V graphs of a fixed resistor, filament lamp and diode	Mission 6 · I–V characteristics
09	Investigating I–V characteristics REQUIRED PRACTICAL Required practical: measure the I–V characteristics of a resistor, lamp and diode	Required practical · I–V characteristics
10	Thermistors, LDRs and sensors How thermistors and LDRs respond to temperature and light, and sensor circuits	Mission 6 · Thermistor and LDR
11	Power and energy transfer Power, $E = P t$, $P = V I$, $P = I^2 R$, energy transfers in appliances	Mission 7 · Power & the National Grid
12	Mains, safety and the National Grid ac and dc, 230 V 50 Hz mains, three-core cable, earthing, the National Grid	Missions 7 and 8

Companion website: panphy.app/gcsephy/year10phy/unit01 · missions, interactive equation triangles, required-practical pages and the Exam Zone.

EQUATION TOOLKIT

The equations you have met so far. You will meet these one at a time. Learn each equation **in words** as well as symbols: your exams may or may not give you an equation sheet, and knowing them by heart makes every question faster.

$$Q = I t$$

charge flow = current $\times$ time

C, A, s

Rearranged: $I = Q \div t$ $t = Q \div I$

Lesson 2

$$E = Q V$$

energy transferred = charge $\times$ p.d.

J, C, V

Rearranged: $V = E \div Q$ $Q = E \div V$

Lesson 3

$$V = I R$$

p.d. = current $\times$ resistance

V, A, Ω

Rearranged: $I = V \div R$ $R = V \div I$

Lesson 4

$$R_{\text{total}} = R_1 + R_2$$

total resistance in series

Ω

Rearranged: (series only)

Lesson 6

FOUR STEPS FOR EVERY CALCULATION

1

Write down what you know, with units. Convert units first (minutes $\rightarrow$ seconds, kW $\rightarrow$ W, mA $\rightarrow$ A).

2

Write the equation you will use, in symbols.

3

Substitute the numbers, then rearrange if needed.

4

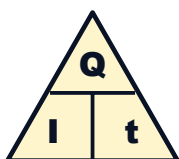
Give the answer with a unit. Ask: is it sensible?

CONVERT	HOW	EXAMPLE
minutes $\rightarrow$ seconds	$\times 60$	3.0 min = 180 s
hours $\rightarrow$ seconds	$\times 3600$	2.0 h = 7200 s
milliamps (mA) $\rightarrow$ amps (A)	$\div 1000$	250 mA = 0.25 A
kilowatts (kW) $\rightarrow$ watts (W) $\cdot$ kilohms (k Ω) $\rightarrow$ ohms (Ω)	$\times 1000$	2.2 kW = 2200 W
kilovolts (kV) $\rightarrow$ volts (V)	$\times 1000$	400 kV = 400 000 V

EQUATION TRIANGLES: HELP WITH REARRANGING

- 1 Cover the quantity you want to find.
- 2 The two left **side by side**? **Multiply** them.
- 3 One **above** the other? **Divide** the top by the bottom.

Example: to find I from $Q = I t$, cover I. Q is above t, so $I = Q \div t$. Triangles only help with rearranging: you still need to learn each equation.

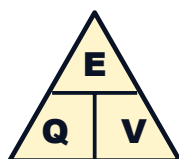


$$Q = I t$$

$$Q = I \times t$$

$$I = Q \div t$$

$$t = Q \div I$$

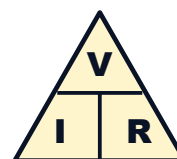


$$E = Q V$$

$$E = Q \times V$$

$$Q = E \div V$$

$$V = E \div Q$$



$$V = I R$$

$$V = I \times R$$

$$I = V \div R$$

$$R = V \div I$$

Interactive version: panphy.app/gcsephy/year10phy/unit01/equation-triangles.

RESISTANCE OF A WIRE

AQA Combined RP 15 · Physics RP 3 · Part A

BIG QUESTION If you double the length of a wire, what happens to its resistance?

BY THE END YOU CAN

- set up a circuit to find resistance from V and I
- identify and control the variables
- record, process and graph results
- write a conclusion using evidence

independent variable

dependent variable

control variable

anomaly

line of best fit

ONLINE COMPANION

Required practical · Resistance

Method, example results and a printable worksheet (Part A).

panphy.app/gcsephy/year10phy/unit01/practical/resistance/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

Calculate R for 1.2 V and 0.40 A.

2

Why does heating change a wire's resistance?

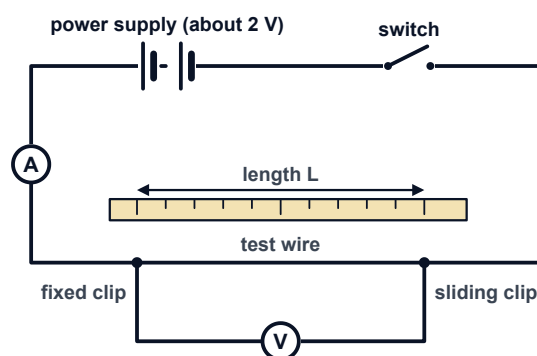
3

Draw the symbols for an ammeter and a voltmeter.



EXPLAIN

PLAN THE PRACTICAL



The wire is taped along a metre ruler. The voltmeter is connected across the wire between the two clips. One crocodile clip stays at 0 cm; you slide the other to change the length. The ammeter measures the current *through* the wire; the voltmeter measures the p.d. *across* the length between the clips. Then $R = V \div I$.

1 Complete the variables table.

Independent variable (what you change)	
Dependent variable (what you find)	
Control variables (keep the same)	

METHOD

- 1 Set up the circuit. Fix one clip at 0 cm.
- 2 Put the sliding clip at 10 cm. Switch on.
- 3 Quickly read the ammeter and voltmeter. Switch off.
- 4 Move the clip to 20 cm and repeat, up to 100 cm.
- 5 Calculate $R = V \div I$ for each length.
- 6 Repeat any reading that does not fit the pattern.

STAY SAFE

- The thin wire can get hot enough to burn. Do not touch it while the current is on.
- Switch off between readings. This also stops the wire heating up, which would change its resistance.
- Keep the p.d. low.



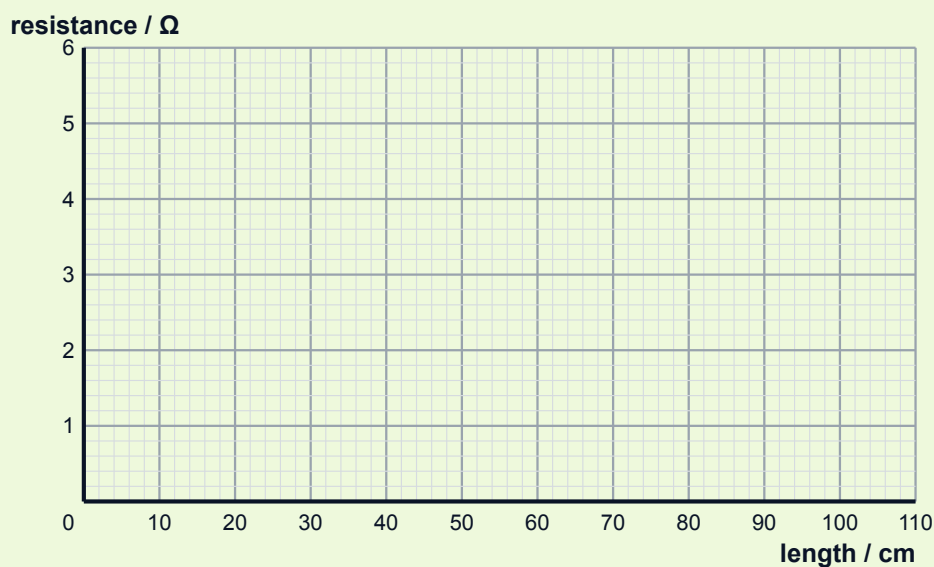
INVESTIGATE RESULTS

Answer edition: example results. Yours will differ.

- 7 Record your readings and calculate the resistance.

Length / cm	p.d. / V	current / A	$R = V \div I / \Omega$
10			
20			
30			
40			
50			
60			
70			
80			
90			
100			

- 8 Plot resistance (up the side) against length (along the bottom). Draw a line of best fit.



PRACTISE ANALYSE AND EVALUATE

- 2 Describe the pattern shown by your graph.

As the length increases, the resistance ...

- 3 Use your graph to find the resistance of 75 cm of the wire.

4 Circle any anomalous result in your table. Suggest one possible cause.

5 Why is the circuit switched off between readings? Tick **one** box.

☐ to save the battery

☐ to stop the wire heating up and changing its resistance

☐ to reset the ammeter

☐ to make the voltmeter more accurate

6 A student's line of best fit crosses the resistance axis a little above zero. Suggest why.



AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: A 6-mark method covers: the circuit, what you change and how, what you measure, how you calculate R , how you keep it fair and safe, and what you do with the results.

9 Describe how you would investigate how the resistance of a wire depends on its length. **[6 marks]**



SUMMARY

- Resistance of a wire: $R = V \div I$, ammeter in series, voltmeter across the wire.
- Change the length. Keep material, diameter and temperature the same.
- Resistance is **directly proportional** to length.
- Switch off between readings so the wire does not heat up.

COVER THE PAGE AND ANSWER

1 What are the independent and dependent variables?

.....

3 How does resistance change with length?

.....

2 Why keep the current small?

.....

4 Name two control variables.

.....

I CAN...

draw and set up the circuit to find a resistance



name the independent, dependent and control variables



plot a graph and draw a line of best fit



spot an anomaly and suggest a cause



SERIES AND PARALLEL CIRCUITS

AQA Combined 6.2.2 · Physics 4.2.2

BIG QUESTION When one bulb in your kitchen blows, why don't all the other lights in the house go out?

BY THE END YOU CAN

- tell series and parallel circuits apart
- use the rules for current and p.d. in each
- calculate total resistance in series
- explain why adding a parallel branch lowers the total resistance

series

parallel

branch

junction

total resistance

ONLINE COMPANION

Mission 5 · Series & parallel

Revision notes, 6 practice questions and a go-further box.

panphy.app/gcsephy/year10phy/unit01/lesson/series-and-parallel/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

0.50 A flows through a $12\ \Omega$ resistor. Calculate the p.d.

2

How does the resistance of a wire depend on its length?

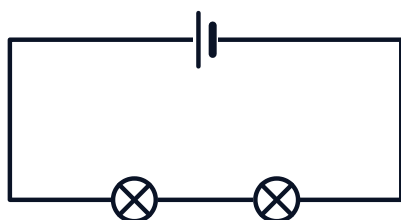
3

What is the unit of p.d., and what does it mean?

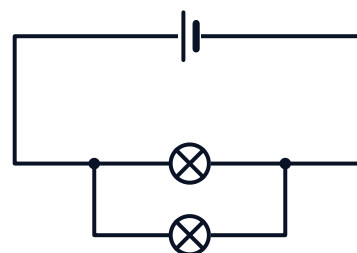


EXPLAIN

TWO WAYS TO CONNECT



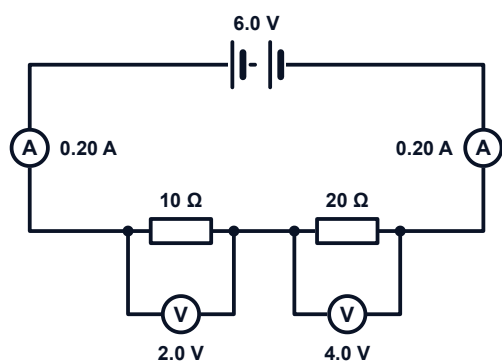
Series: one loop.



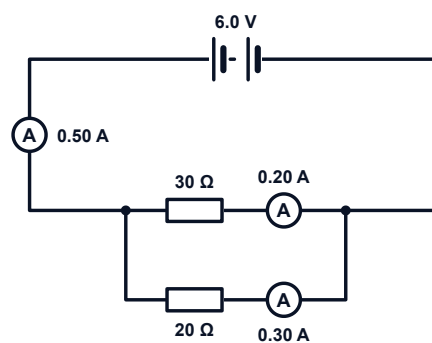
Parallel: two branches.

In **series** there is one loop. In **parallel** the current splits at a **junction** into separate **branches**, and each branch makes its own loop through the cell.

	SERIES (ONE LOOP)	PARALLEL (BRANCHES)
Current	the same everywhere	splits at a junction; branch currents add up to the supply current
p.d.	shared : component p.d.s add up to the supply p.d.	the same across each branch as across the supply
Resistance	adds : $R_{\text{total}} = R_1 + R_2$	total is less than the smallest branch resistance
One lamp breaks	all the lamps go out	the other branches keep working



Series: the same current everywhere; $2.0 \text{ V} + 4.0 \text{ V} = 6.0 \text{ V}$.



each branch has 6.0 V across it

Parallel: $0.20 \text{ A} + 0.30 \text{ A} = 0.50 \text{ A}$ from the battery.

$$R_{\text{total}} = R_1 + R_2$$

total resistance of components in series

all resistances in ohms (Ω)

Adding a resistor in series always increases the total resistance.

WORKED EXAMPLE

A 5.0Ω and a 7.0Ω resistor are in series with a 6.0 V battery. Find the total resistance, the current and the p.d. across each resistor.

TOTAL R $5.0 + 7.0 = 12 \Omega$

CURRENT $I = V \div R = 6.0 \div 12 = 0.50 \text{ A}$

P.D. 5Ω $V = I R = 0.50 \times 5.0 = 2.5 \text{ V}$

P.D. 7Ω $V = 0.50 \times 7.0 = 3.5 \text{ V}$

CHECK $2.5 \text{ V} + 3.5 \text{ V} = 6.0 \text{ V} \checkmark$

YOUR TURN

A 3.0Ω and a 9.0Ω resistor are in series with a 12 V battery. Find the same four things.

TOTAL R

CURRENT

P.D. 3Ω

P.D. 9Ω

CHECK

WHY DOES A PARALLEL BRANCH LOWER THE RESISTANCE?

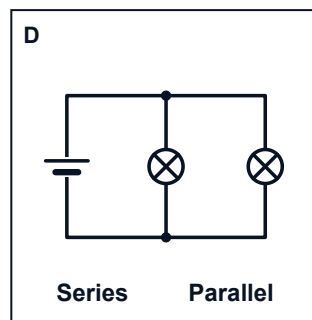
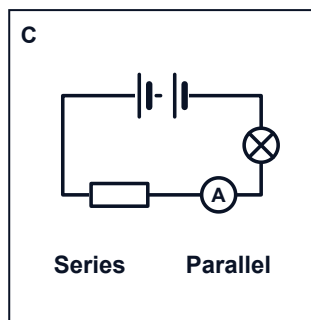
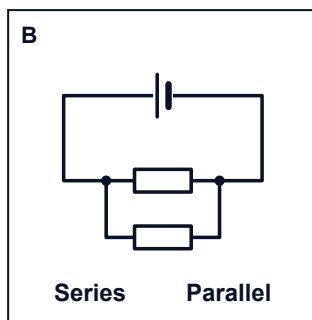
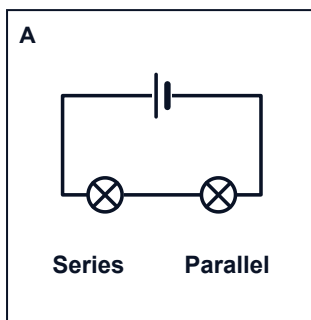
Think of supermarket checkouts. Opening a second checkout does not slow down the first one: it gives shoppers *another route*, so more people get through each minute. A parallel branch is another route for charge. The supply current goes up at the same p.d., so the total resistance ($V \div I$) goes down.

Where the model breaks: shoppers choose a queue; charge divides according to each branch's resistance.

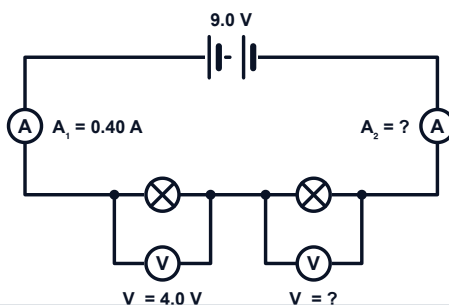


PRACTISE
TRY IT

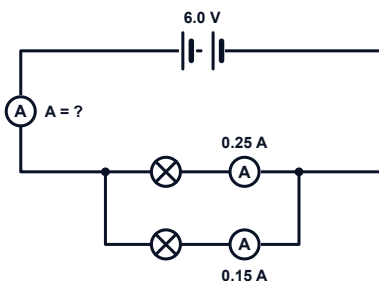
- 1 Is each circuit series or parallel? Circle one.



- 2 Series circuit: write down the readings on A_2 and V_2 . Give a reason for each.



- 3 Parallel circuit: what does ammeter A read? What is the p.d. across each lamp?



- 4 **SUPPORTED** A 10Ω and a 20Ω resistor are in series with a 6.0 V battery. Calculate the total resistance and the current.

TOTAL R	
EQUATION	
SUBSTITUTE	
ANSWER	unit:

- 5 A 6.0Ω and a 12Ω resistor are connected in **parallel**. Which could be their total resistance?

☐ 18Ω

☐ 9.0Ω

☐ 6.0Ω

☐ 4.0Ω

6 Answer the big question. Give **two** reasons why the lights in a house are wired in parallel.



AQA-STYLE
EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: Show the equation for each step. You can still get later marks with an earlier wrong answer if your method is right.

7 A $4.0\ \Omega$ resistor and an $8.0\ \Omega$ resistor are connected in series to a 12 V battery. (a) Calculate the total resistance. (b) Calculate the current. (c) Calculate the p.d. across the $8.0\ \Omega$ resistor. **[5 marks]**



REVIEW LATER
REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Series: **same current**; p.d.s **add up**; $R_{\text{total}} = R_1 + R_2$.
- Parallel: **same p.d.** across each branch; branch currents **add up**.
- Adding a branch in parallel **lowers** the total resistance: there is another path, so more current flows.
- Houses are wired in parallel.

COVER THE PAGE AND ANSWER

1 In series, what is the same everywhere?

2 In parallel, what is the same for each branch?

3 Total resistance of $3\ \Omega$ and $5\ \Omega$ in series?

4 Branch currents $0.2\ \text{A}$ and $0.3\ \text{A}$. Supply current?

I CAN...

tell series and parallel circuits apart



use the current and p.d. rules in series and parallel



calculate total resistance, current and p.d.s in a series circuit



explain why parallel branches reduce the total resistance



RESISTORS IN SERIES AND PARALLEL

AQA Combined RP 15 · Physics RP 3 · Part B

BIG QUESTION Two identical resistors, three ways to connect them. Which arrangement lets the most current flow?

BY THE END YOU CAN

- measure the total resistance of resistors in series and in parallel
- compare results with $R_{\text{total}} = R_1 + R_2$
- explain the results using paths and current
- evaluate the measurements

total resistance

prediction

uncertainty

resolution

evaluate

ONLINE COMPANION

Required practical · Resistance (Part B)

Method, example results and a printable worksheet (Part B).

panphy.app/gcsephy/year10phy/unit01/practical/resistance/



RETRIEVE

DO NOW

From earlier lessons. No notes!

1

Total resistance of $10\ \Omega$ and $15\ \Omega$ in series?

2

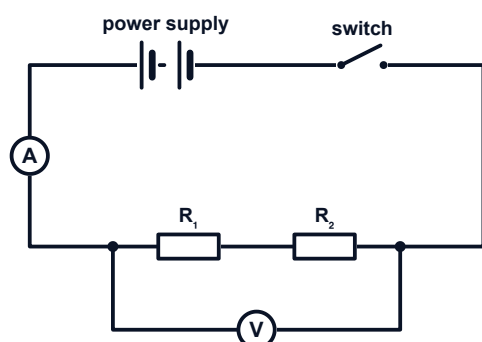
In parallel, what is the same for each branch?

3

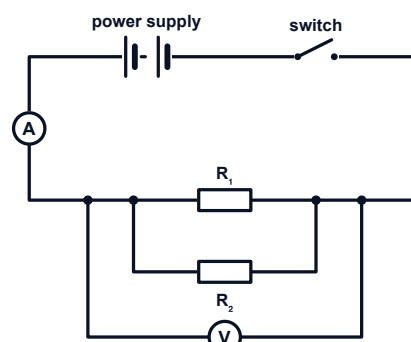
The supply current is $0.9\ \text{A}$. One branch carries $0.4\ \text{A}$. What does the other carry?



EXPLAIN

PLAN THE PRACTICAL

Series: the voltmeter goes across both resistors.



Parallel: the voltmeter goes across the pair.

Measure the p.d. **across the whole combination** and the current **from the supply**. Then the total resistance is $R = V \div I$.

- 1 Connect R_1 on its own. Measure V and I . Repeat for R_2 on its own.
- 2 Connect R_1 and R_2 in series. Measure V across both, and I .

- 3 Connect R_1 and R_2 in parallel. Measure V across the pair, and the supply current.
- 4 Calculate $R = V \div I$ for each arrangement.

STAY SAFE

- Resistors can get hot. Switch off between readings.
- Keep the p.d. low, and check the circuit before switching on.

- 1 **Predict** before you measure. Rank R_1 alone, series and parallel from the *largest* to the *smallest* total resistance.



INVESTIGATE RESULTS

Answer edition: example results with two $10\ \Omega$ resistors at 3.0 V .

- 6 Record your readings.

Arrangement	p.d. / V	current / A	$R = V \div I / \Omega$
R_1 on its own			
R_2 on its own			
R_1 and R_2 in series			
R_1 and R_2 in parallel			



PRACTISE ANALYSE AND EVALUATE

- 2 Compare your series result with $R_1 + R_2$. Do they agree?
- 3 Explain why the total resistance in parallel is *less* than either resistor on its own.
Adding a second branch gives the charge ...
- 4 Suggest one source of error in the measurements and one way to reduce it.

- 5 **EXTRA PRACTICE** Resistors of $5.0\ \Omega$, $10\ \Omega$ and $15\ \Omega$ are connected in series with a $6.0\ \text{V}$ supply. Calculate the current.

TOTAL R	
EQUATION	
SUBSTITUTE	
ANSWER	unit:



AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: 'Explain' needs a chain: another path $\rightarrow$ more current at the same p.d. $\rightarrow$ lower total resistance.

- 7 Two identical resistors each have a resistance of $12\ \Omega$. (a) Calculate their total resistance in series. (b) A student measures $6.0\ \Omega$ when they are connected in parallel. Explain why this is less than $12\ \Omega$. (c) State one variable the student should keep the same for a fair comparison. **[5 marks]**



REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Find total resistance from the **supply current** and the **p.d. across the combination**: $R = V \div I$.
- Series: the total is the sum ($R_1 + R_2$): bigger than either.
- Parallel: the total is smaller than either resistor.
- Evaluate by naming a specific source of error and a specific improvement.

COVER THE PAGE AND ANSWER

- | | |
|---|---|
| 1 How do you find the total resistance of a combination in the lab? | 2 Two $10\ \Omega$ resistors in series? |
| 3 Is two $10\ \Omega$ in parallel more or less than $10\ \Omega$? | 4 Give one source of error in this practical. |

I CAN...

- | | |
|---|---|
| set up series and parallel combinations with the meters in the right places | ☐ ☐ ☐ |
| calculate total resistance from V and I | ☐ ☐ ☐ |
| explain why series adds and parallel lowers resistance | ☐ ☐ ☐ |
| suggest a specific improvement to a practical | ☐ ☐ ☐ |

KEY WORDS

Learn the meaning and spelling. In exams, precise words earn marks: say “p.d. across” and “current through”.

Ammeter · Measures current. Connected in series.

Ampere (A) · Unit of current. $1\text{ A} = 1\text{ C per second}$.

Anomaly · A result that does not fit the pattern of the others.

Battery · Two or more cells joined together.

Charge (Q) · Carried by particles such as electrons. Unit: coulomb (C).

Conventional current · The agreed direction of current: from + to -.

Current (I) · The rate of flow of charge.

Directly proportional · If one quantity doubles, the other doubles. The graph is a straight line through the origin.

Electron · A tiny negatively charged particle. Free electrons carry the current in metals.

Ohm (Ω) · Unit of resistance.

Ohmic conductor · Current is directly proportional to p.d. at constant temperature.

Parallel · Components on separate branches. A voltmeter is connected in parallel.

Potential difference (V) · Energy transferred per unit charge between two points. Unit: volt (V).

Resistance (R) · How much a component opposes the current. Unit: ohm (Ω).

Series · Components connected one after another in a single loop.

Volt (V) · Unit of p.d. $1\text{ V} = 1\text{ J per C}$.

Voltmeter · Measures p.d. Connected in parallel, across a component.

FINISH

PROGRESS TRACKER

Colour one circle for each statement: **red** = not yet, **amber** = getting there, **green** = I can do this without help. Revisit your reds first.

I CAN...	R · A · G
05 · Resistance of a wire	
draw and set up the circuit to find a resistance	
name the independent, dependent and control variables	
plot a graph and draw a line of best fit	
spot an anomaly and suggest a cause	
06 · Series and parallel circuits	
tell series and parallel circuits apart	
use the current and p.d. rules in series and parallel	
calculate total resistance, current and p.d.s in a series circuit	
explain why parallel branches reduce the total resistance	
07 · Resistors in series and parallel	
set up series and parallel combinations with the meters in the right places	
calculate total resistance from V and I	
explain why series adds and parallel lowers resistance	
suggest a specific improvement to a practical	

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising.

Answers to the longer questions are in your teacher's answer edition.

LESSON 5 · RESISTANCE OF A WIRE

Do now: (1) $R = 1.2 \div 0.40 = 3.0 \, \Omega$ (2) Ions vibrate more, so more collisions: R increases. (3) Circles containing A and V .

Cover the page: (1) Length; resistance. (2) To stop the wire heating (R would increase). (3) Directly proportional. (4) Material; diameter (thickness).

LESSON 6 · SERIES AND PARALLEL CIRCUITS

Do now: (1) $V = 0.50 \times 12 = 6.0 \, V$ (2) Directly proportional. (3) Volt; joules per coulomb.

Cover the page: (1) The current. (2) The p.d. (3) $8 \, \Omega$ (4) $0.5 \, A$

LESSON 7 · RESISTORS IN SERIES AND PARALLEL

Do now: (1) $25 \, \Omega$ (2) The p.d. (3) $0.5 \, A$

Cover the page: (1) $R = V \div I$, with V across the combination and I from the supply. (2) $20 \, \Omega$ (3) Less (it is $5 \, \Omega$). (4) Resistance of leads; meter resolution; heating.